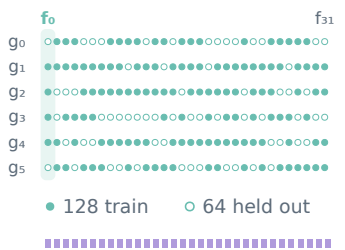
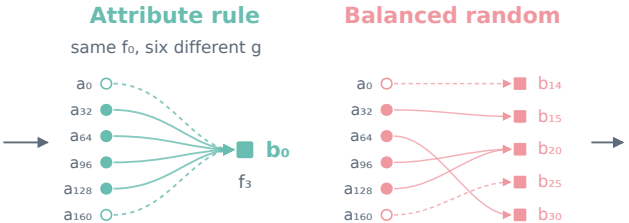
A Attribute-defined object prediction

01 Assign categorical attributes



02 Create two matched relations



03 Repeat & shuffle sentences

Attributes · all 224 entities ×128

x a_0 f_0 g_0 x

x b_0 f_3 g_0 x

Attribute rule · 128 triples ×16

x a_{32} r_0 b_0 x

Random control · 128 triples ×16

x a_{32} r_1 b_{15} x

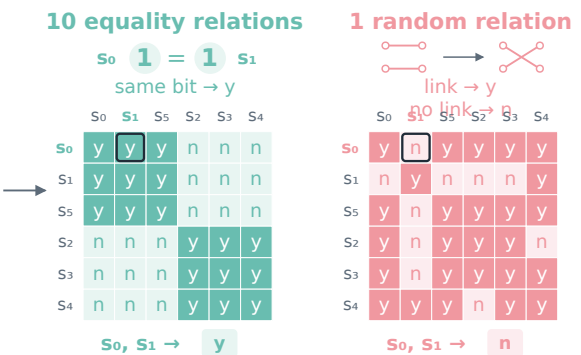
B Attribute equality & symmetric lookup

64 entities → binary pair labels

01 Assign each entity 10 bits

6 of 64 entities shown

02 Label every entity pair



r_0 shown

31 links + self per entity

03 Write training sentences

Attributes · forward + inverse

S_0 $attr_0$ → 1

$attr_0$ 1 → S_0

Same attribute

r_0 S_0 S_1 → y

Random lookup

r_{10} S_0 S_1 → n

Joint training after the equality warm-up

10 equality relations 1 random relation

Random: 10× each equality relation